

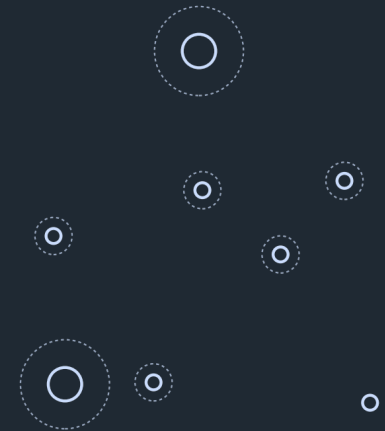
# CrossMEP: A Tiered Synthetic Dataset of Multi-Trade MEP Cross-Sections

A support-design problem distribution for learning-based structural support assembly synthesis

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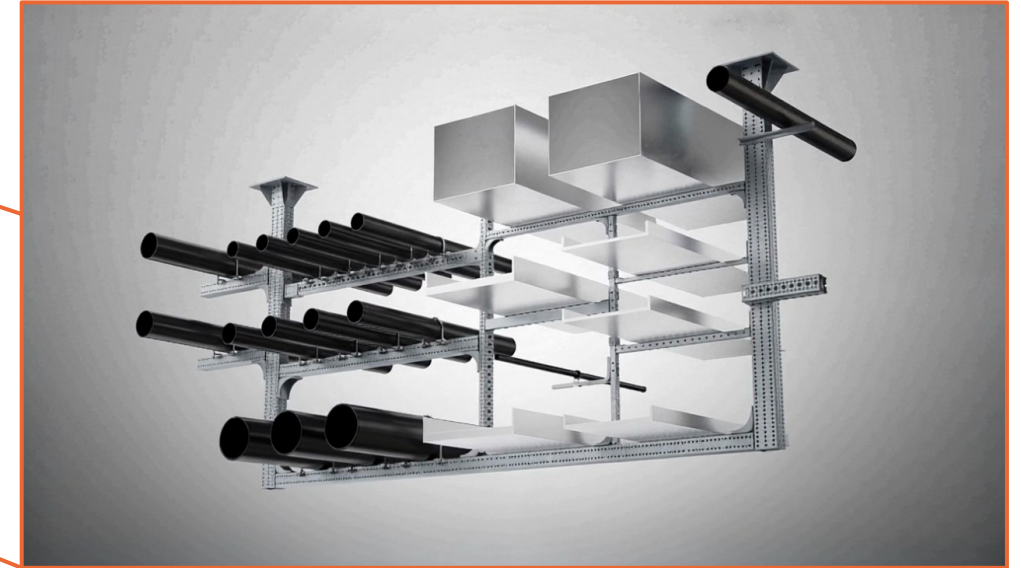


a generated C8 context: eight services on three rows

# A hospital needs about 10,000 support assemblies, each designed by hand



1 unit = 1 structural support assembly (SSA)



**$\approx 10,000$**

support assemblies in one 200,000 sq ft hospital

**20 min – 2 h**

of engineering for each assembly, designed one by one

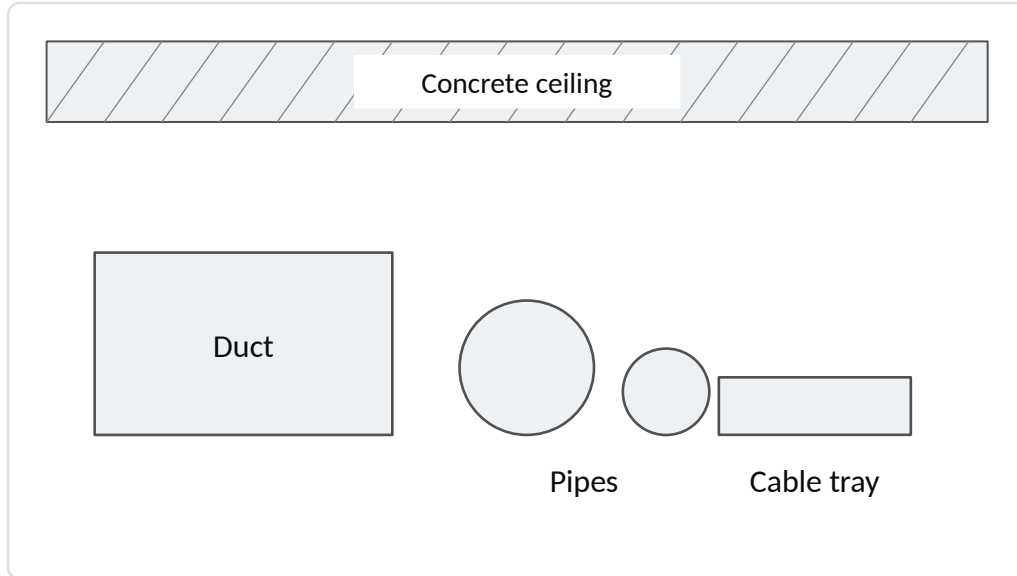
**$\approx \frac{1}{4}$**

of MEP design effort; on the order of \$600K per project

Approximate practitioner estimates.

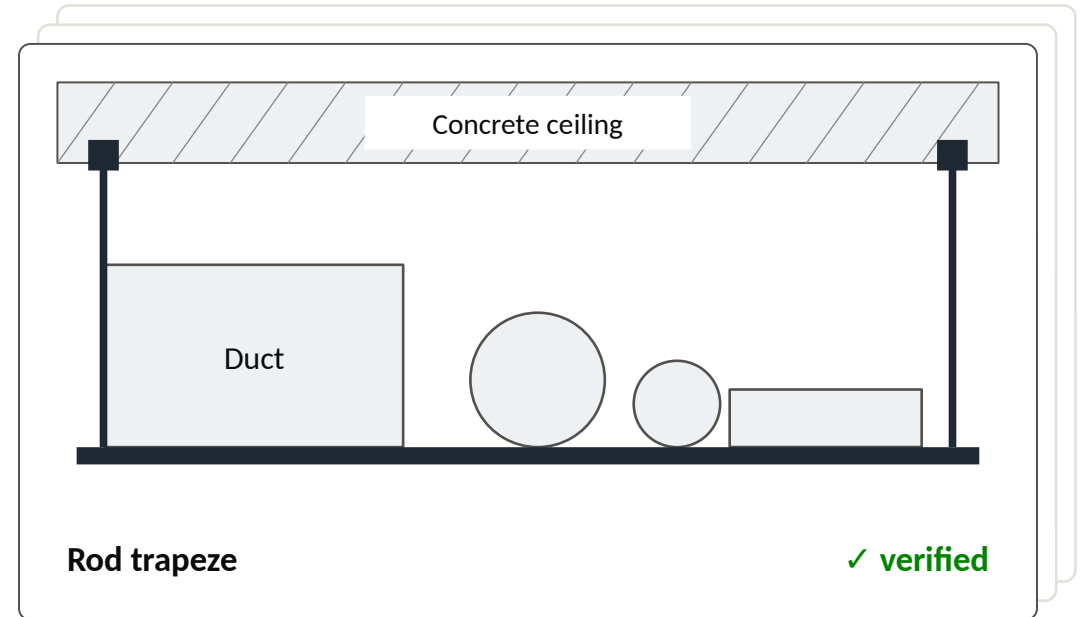
# A designer turns one cross-section into verified support assemblies

In: one cross-section = a context



Structure: ceiling, floor or wall, with anchor zones  
Services: trade, position, size and weight per metre

Out: verified support designs = assemblies



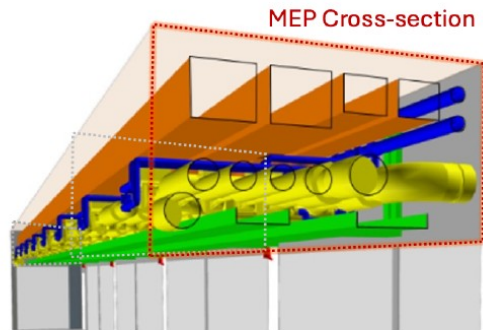
Up to ten designs, ranked by installed cost  
Each checked: statics, anchors, connectors, buildability

**Context, the brief:** the section at one hanger, its structure and services.  
This is what CrossMEP contains.

**Assembly, the answer:** catalog parts that carry the services to the structure. Not in CrossMEP: a method produces it.

The synthesis problem as defined at ISARC 2026.

# Synthesizing the assembly from a catalog and a context has resisted automation



(a) Full MEP route



(b) One CrossMEP context

Design starts after coordination: at each hanger the designer works from the section across the run, not from the whole model.

## Where practice stands

Tools coordinate the model and check a design. Choosing the topology and the parts, from a fixed catalog, is still done by hand, one section at a time.

### 1. Geometric judgment

Engineers choose an assembly's topology from experience.

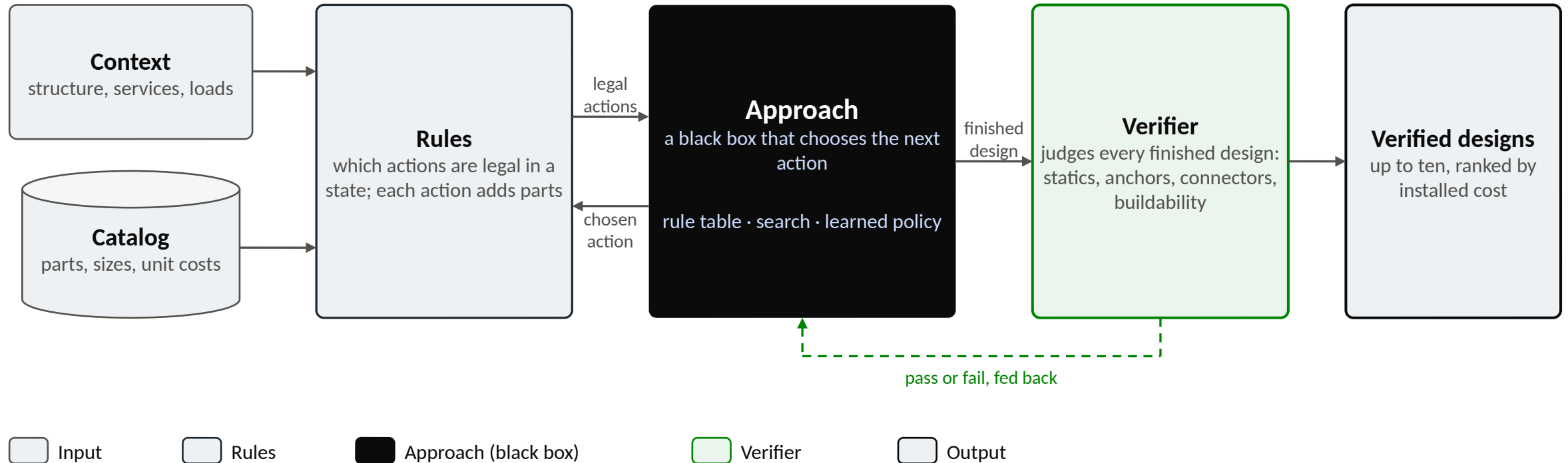
### 2. Rule explosion

Code, load and material constraints interact across thousands of combinations.

### 3. Catalog volatility

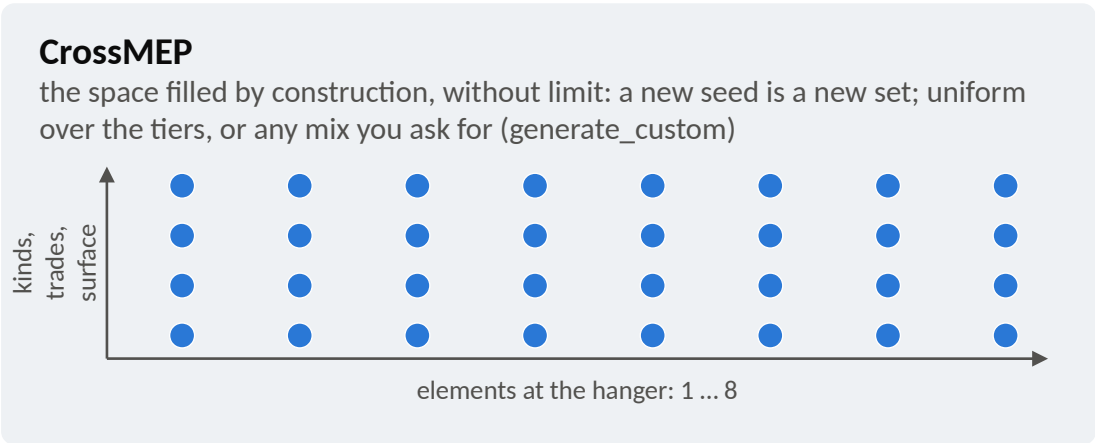
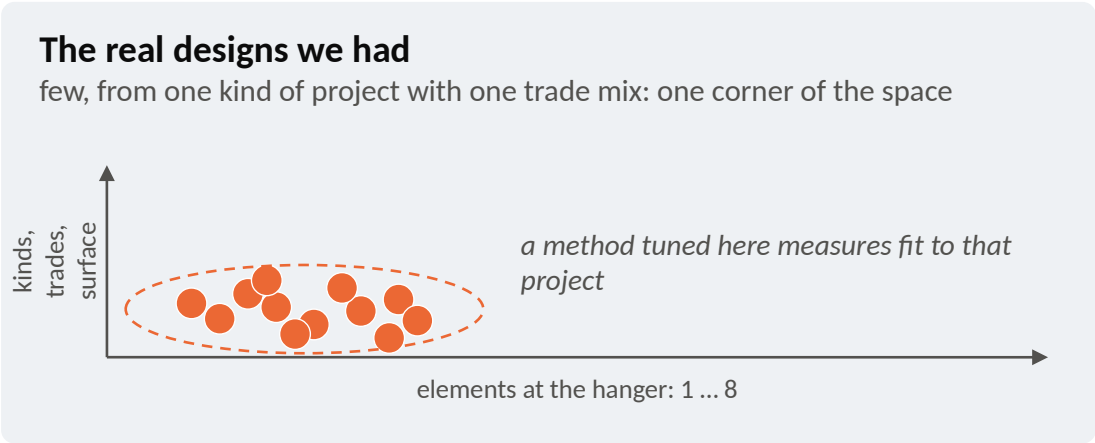
Catalogs change faster than rule-based systems can be rewritten.

# Whatever the method, it needs problems to learn from and a fixed set to compare on



Whatever sits in the black box, it needs **problems to learn from**, a **fixed set to compare on** and **problems that look like practice**.

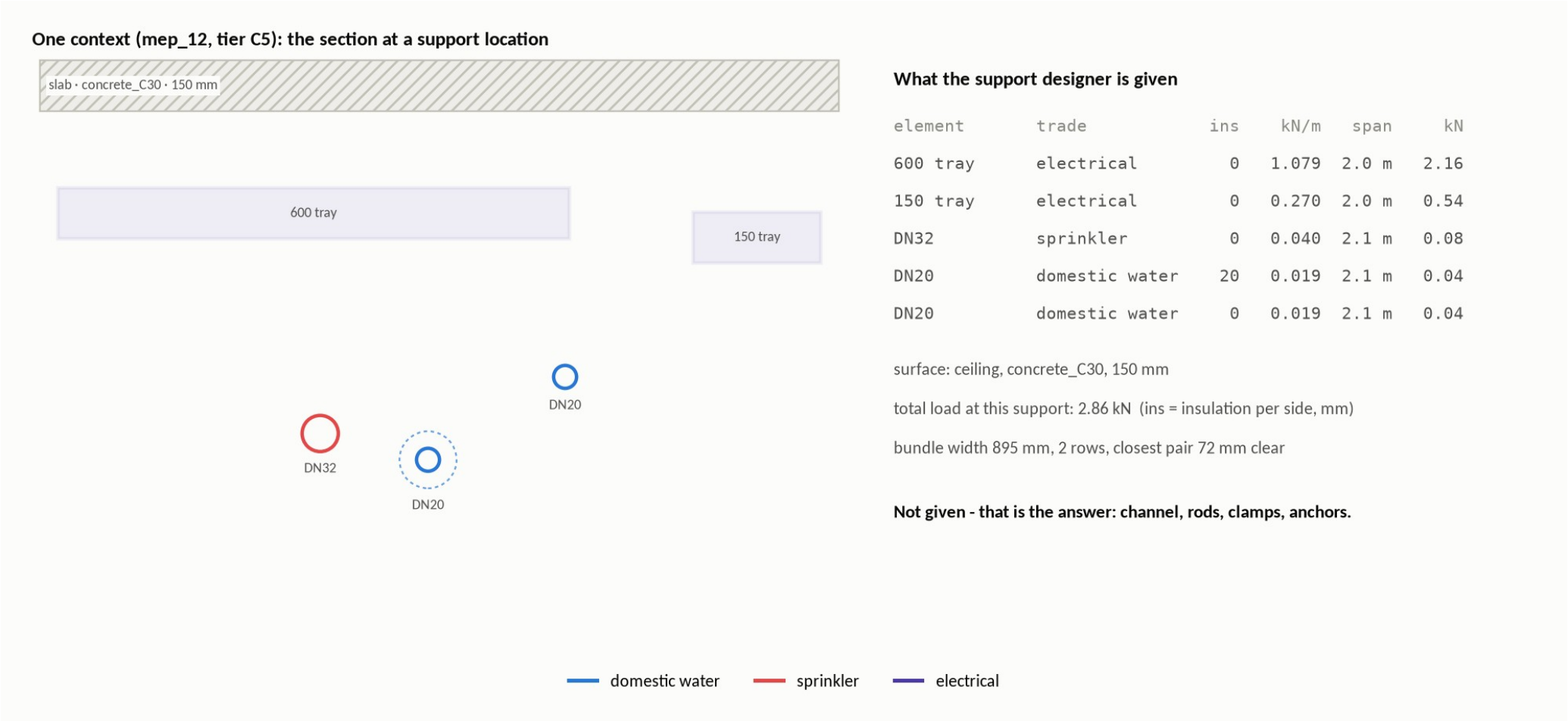
# Real designs cover one corner of the design space; CrossMEP covers all of it



## CrossMEP at a glance

|                                                                               |                                                                               |                                                                            |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>7,000</b><br>contexts in four splits on disjoint seeds                     | <b>31,484</b><br>elements: pipes, cable trays, ducts and conduits; six trades | <b>C1-C8</b><br>eight difficulty tiers by exact element count              |
| <b>0 labels</b><br>the brief, not the answer: no support-assembly information | <b>Traced</b><br>every constant with its source, or declared a design choice  | <b>Open</b><br>data CC BY 4.0, code MIT; a seeded, deterministic generator |

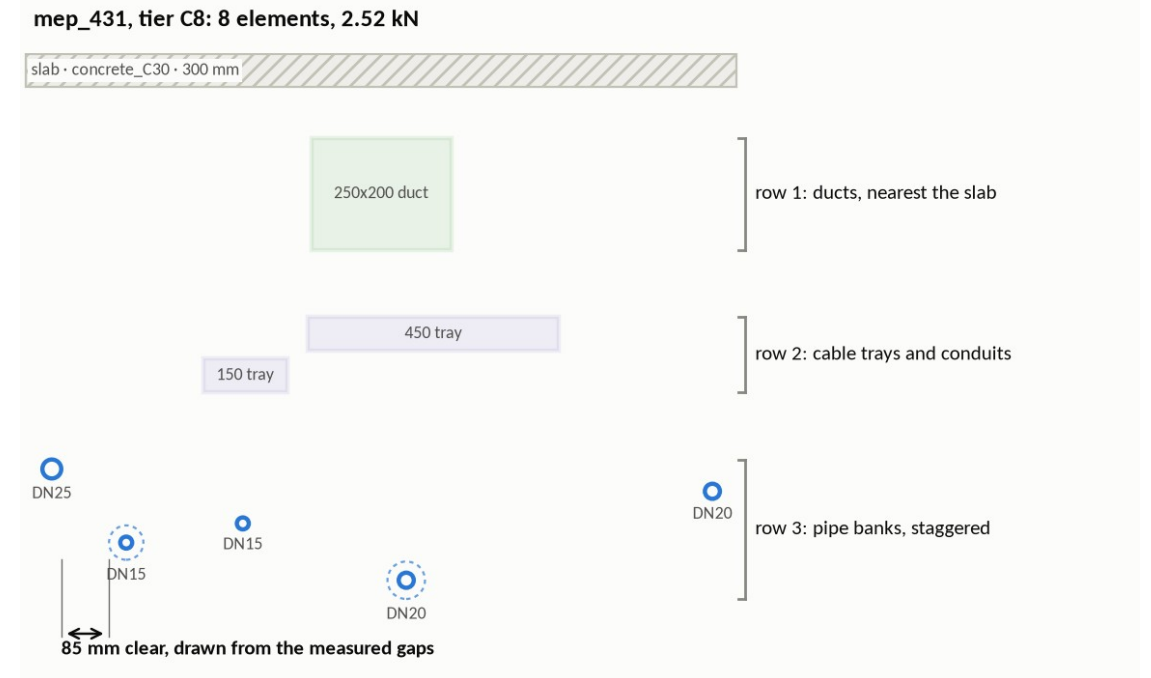
# We generate what a designer receives: the section at one hanger



Per element: kind, service and trade · bare size · insulation per side · load per metre × the span it was sized at = load at the support · position along and out from the surface. Per context: slab or wall, substrate, thickness. Absent by design: channel, rods, clamps, anchors — and any “correct” answer, since a feasible support depends on the catalog you build from.

# Sections follow trade practice and measured spacing, not random shapes

- 1 Tier.** Cn fixes the exact element count,  $n = 1 \dots 8$ .
- 2 Surface.** Ceiling or wall (78 / 22 %); concrete slab or wall, 150–300 mm thick.
- 3 Services.** Grouped as trades run: hot + cold pairs, flow + return, chilled banks, sprinkler mains, 1–2 trays, ducts, conduit groups of 2–6.
- 4 Rows.** Bulky first: ducts nearest the surface, then trays and conduits, then pipes.
- 5 Spacing.** Gaps drawn from gaps measured on a built project (at least 25 mm); stagger within rows; electrical above wet on walls.
- 6 Check.** Every pair at least 25 mm clear; rows centred; sizes, insulation, loads and positions recorded.



One generated benchmark context, three rows by priority; the gap marked is one draw from the measured distribution. Every context is seeded: the same seed gives the same file, byte for byte.

# Sizes, spans and loads come from standards; the choices are declared

| Element                    | Sizes                                                          | Load basis                                                            |
|----------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Pipes DN15–100</b>      | EN 10220 / EN 10255 medium series                              | steel + water × ASME B31.1 water-service span (published points only) |
| <b>Insulation</b>          | GEG Anlage 8 (heated); 30/50 mm condensation control (chilled) | —                                                                     |
| <b>Cable trays 150–600</b> | IEC 61537 systems, NEMA VE 1 widths                            | full tray: 50 kg/m at 300 mm datum, × 2.0 m (NEMA VE 1 class span)    |
| <b>Ducts</b>               | EN 1505 rectangular; EN 1506 round                             | manufacturer duct-weight table incl. flanges, × 2.4 m (SMACNA)        |
| <b>Conduits Ø20–50</b>     | IEC 61386-1, parallel groups of 2–6                            | steel tube + 40 %-of-bore cable fill, × 2.0 m (IET / BS 7671)         |

## Layout conventions

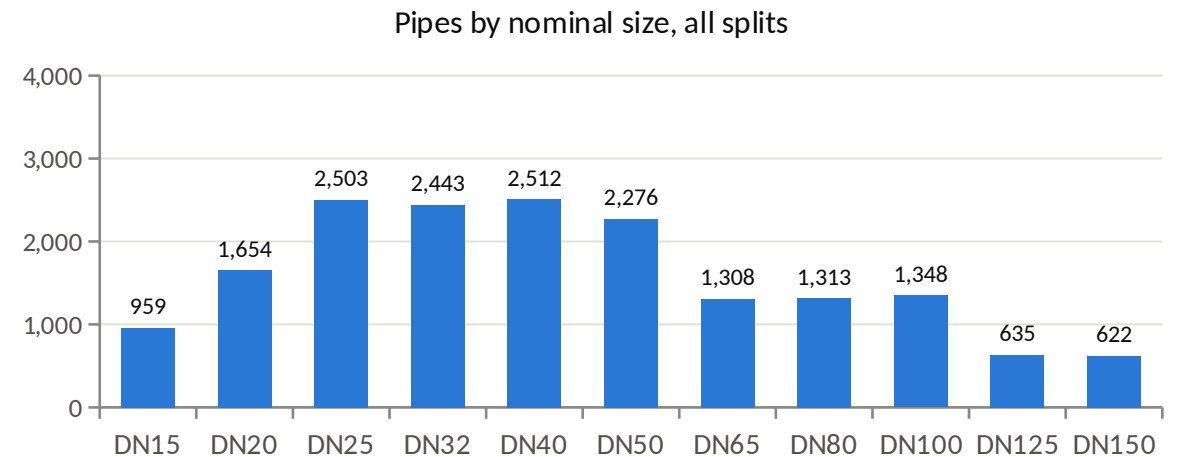
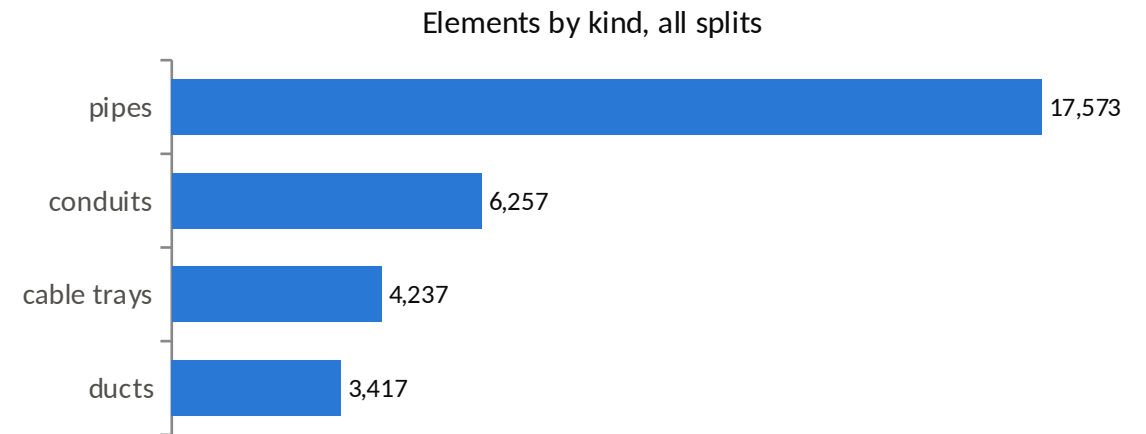
- Bulky services nearest the slab: ducts, then containment, then pipes
- Trades run in banks: hot + cold pairs, flow + return, conduit groups
- Electrical kept above wet services on walls (drip; BS 7671 528.3.2)
- Clear gaps drawn from a distribution measured on a built project; 25 mm minimum
- Within-row stagger calibrated to measured elevation spread

Three kinds of ground: standards (what each element is and weighs), measured open buildings (how close neighbours sit) and declared design choices (trade mix, surface mix, number of rows). Every constant carries its source and status in the repository, and the log says why each standard was chosen over its alternatives.

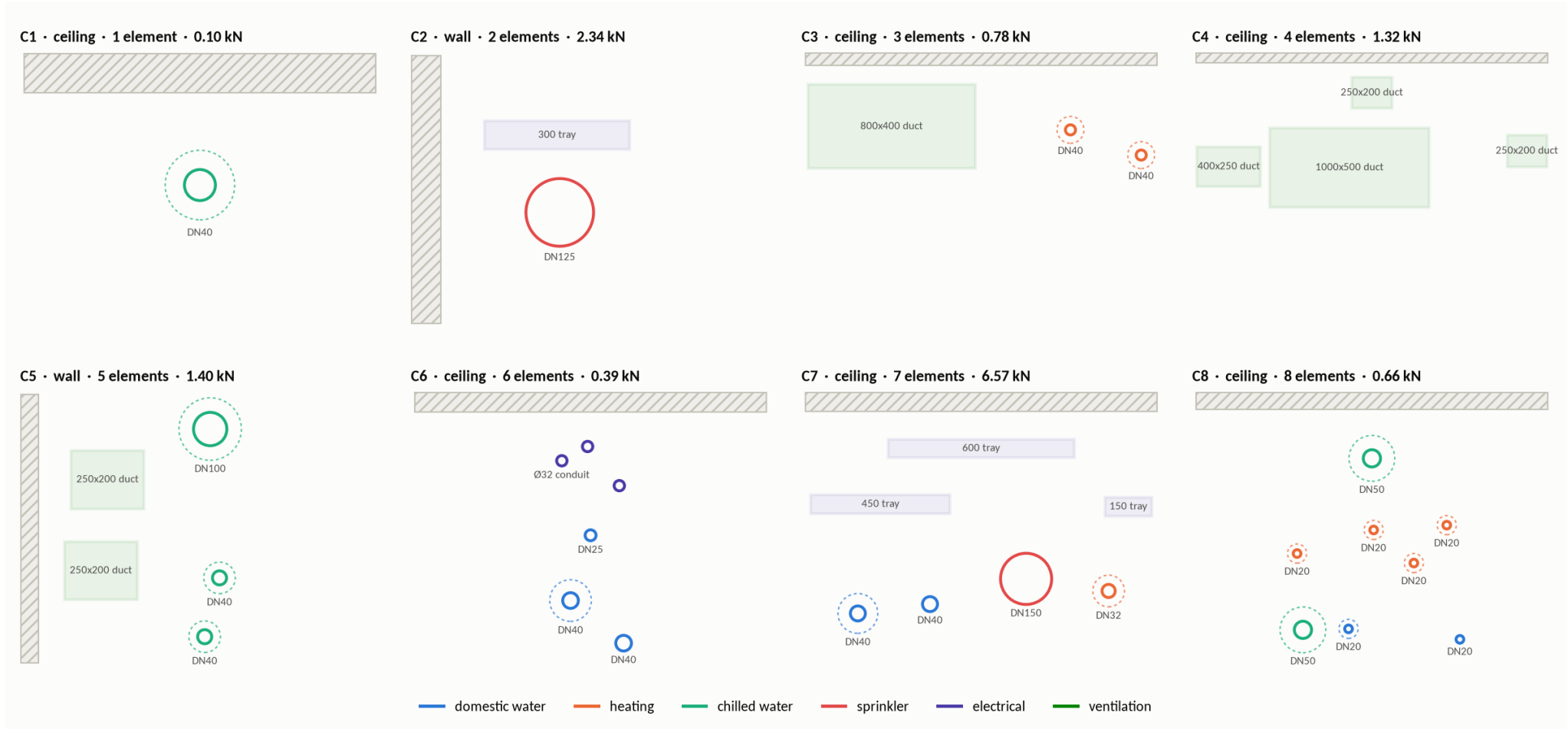
# 7,000 contexts in four splits; the test seeds are never trained on

| split        | contexts     | elements      | seed |
|--------------|--------------|---------------|------|
| train        | 5,000        | 22,500        | 1000 |
| validation   | 500          | 2,242         | 2000 |
| test         | 500          | 2,242         | 3000 |
| benchmark    | 1,000        | 4,500         | 42   |
| <b>total</b> | <b>7,000</b> | <b>31,484</b> |      |

- Tiers assigned round-robin: 625 per tier in train, 125 in the benchmark
- Four in five contexts hang from a ceiling, one in five from a wall
- Six trades: electrical 10,494, domestic water 5,725, heating 5,788, chilled water 3,987, sprinkler 2,073, ventilation 3,417 elements
- Plain JSON, with a JSON Schema, Croissant metadata and a datasheet

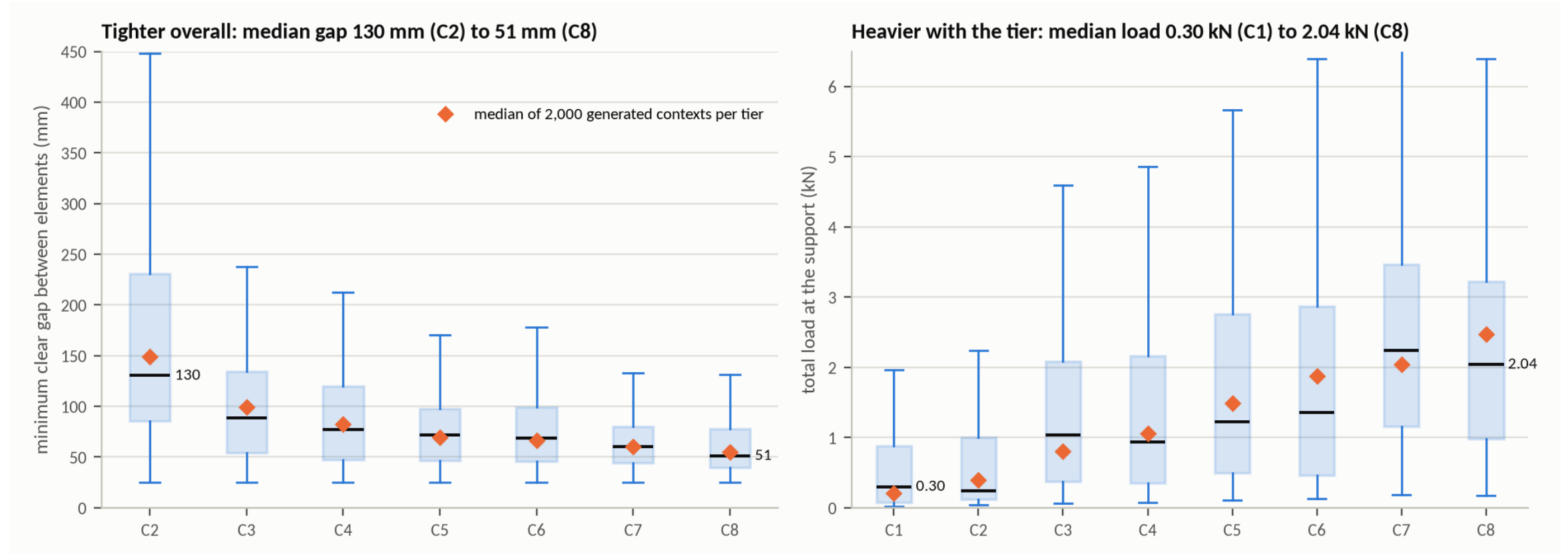


# Difficulty is one number: tier Cn holds exactly n elements



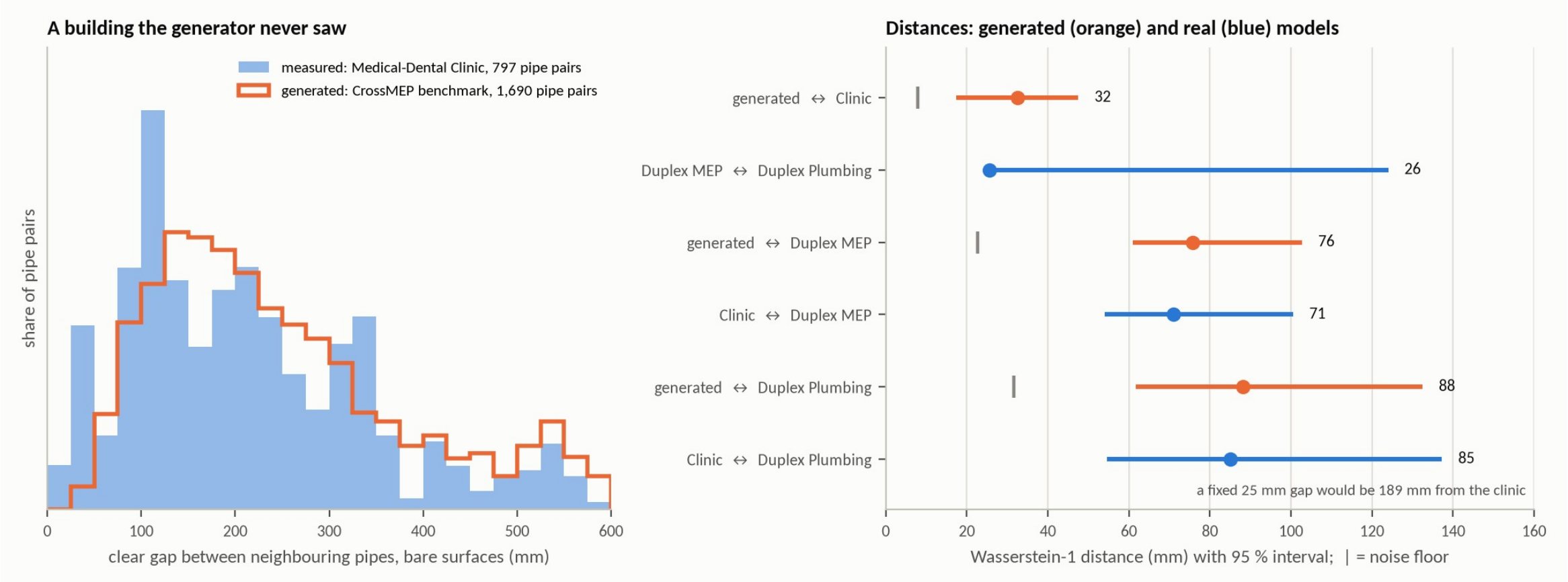
One benchmark context per tier; dashed rings are insulation. Within a tier, kinds, trades, services, surfaces and stacking all vary, so element count is the one controlled difficulty axis. C1 is the most common support in any building; C8 is a congested rack.

# Higher tiers are heavier and tighter, as designed



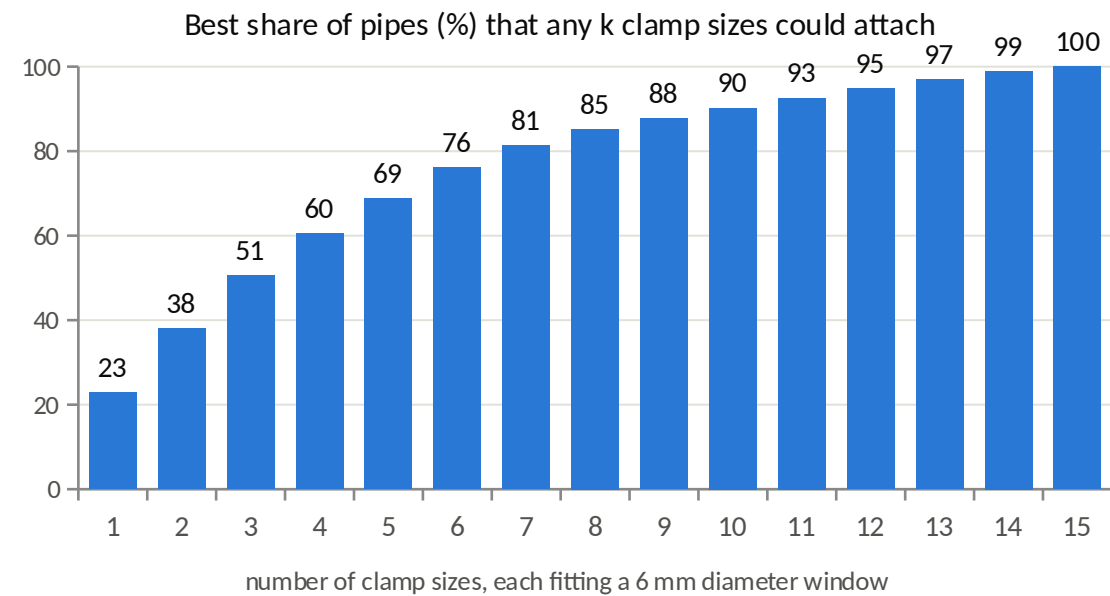
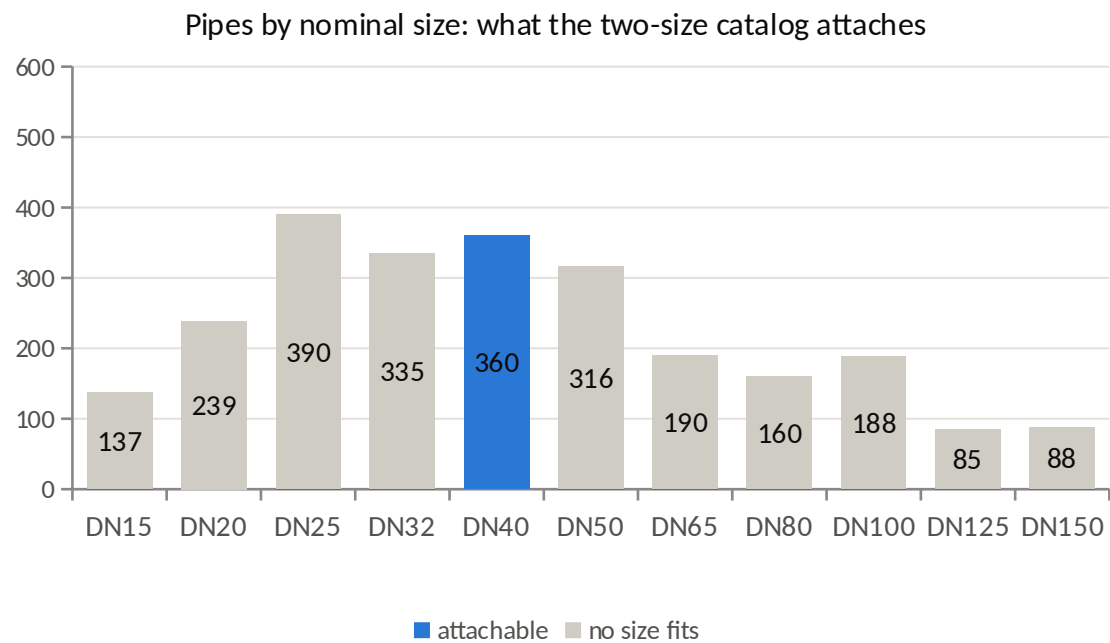
A check that the dataset behaves as designed: the tier fixes the element count; load and congestion follow from the rules. Boxes: benchmark, 125 contexts per tier (IQR, whiskers 1.5 IQR); diamonds: medians of 2,000 generated contexts per tier. Load rises with the tier (rank correlation +0.46; the population medians rise at every step) and the closest gap narrows at every step (-0.38; 149 → 55 mm in the population).

# Spacing holds up on a clinic the generator never saw



The gap distribution was fitted on the duplex; the clinic was held out. Clear gaps between side-by-side pipes, on sections every 250 mm. Clinic (797 pipe pairs): 32 mm from the generated gaps (95 % CI 17–48), more than the 8 mm a perfect generator would show, and about as close as the duplex’s own two models are to each other (26 mm). Models: buildingSMART Duplex Apartment and Medical-Dental Clinic, CC BY 4.0.

# A two-size catalog attaches 1 pipe in 7; the dataset shows what to cover



14.5 %

of benchmark pipes attachable with the paper's two sizes (95 % CI 12.7–16.2); every one is DN40

38.1 %

with two sizes placed where the pipes are; 76 % with six, and every pipe with 15

0 of 2,012

trays, ducts and conduits: a pipe-clamp catalog defines no attachment. Load never binds (1.97 vs 2.5 kN)

# Set the parameters and the generator returns the designer's brief

SERVICES AT THE HANGER

TIER = NUMBER OF SERVICES

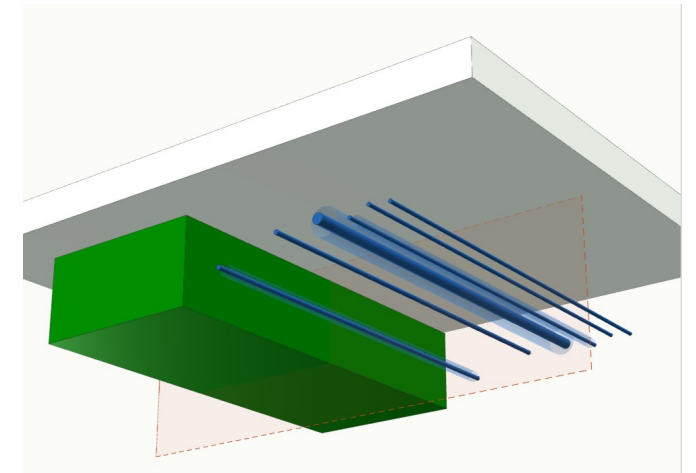
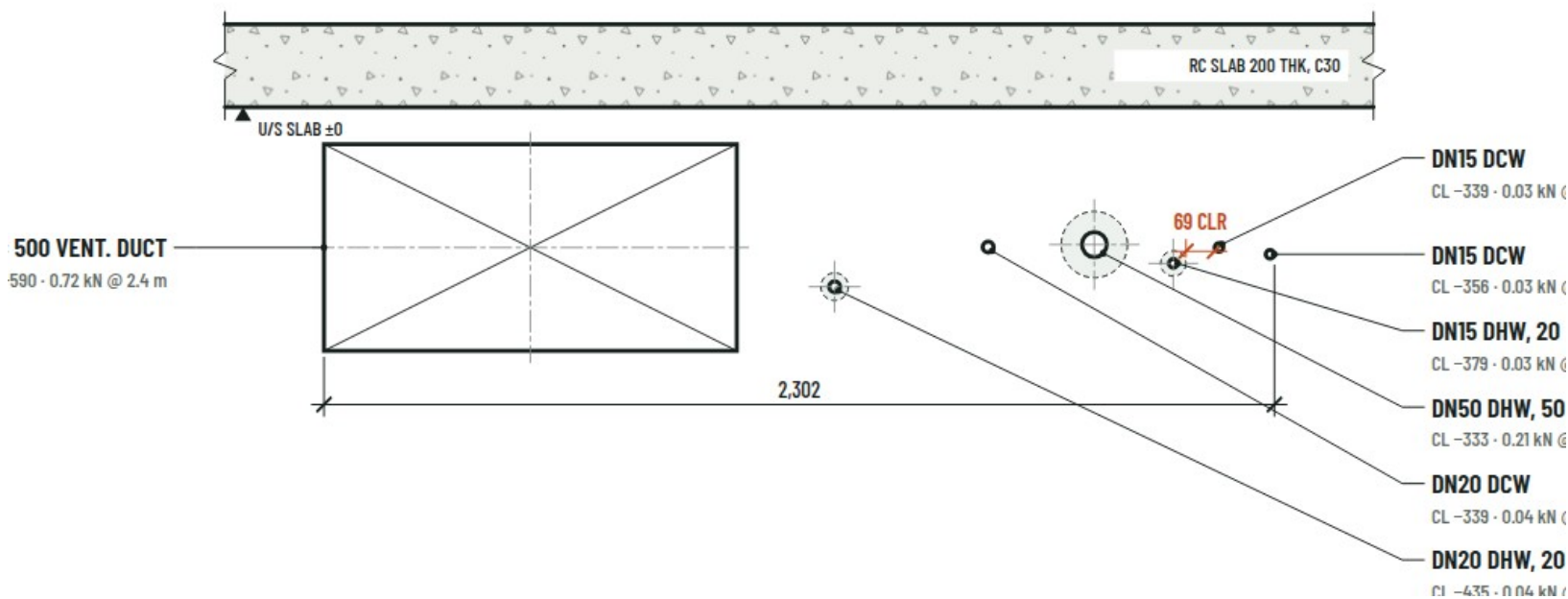
SEED

By tier Exact mix

C1 C2 C3 C4 C5 C6 C7 C8

< 0 >

Generate another



Scan to open the studio

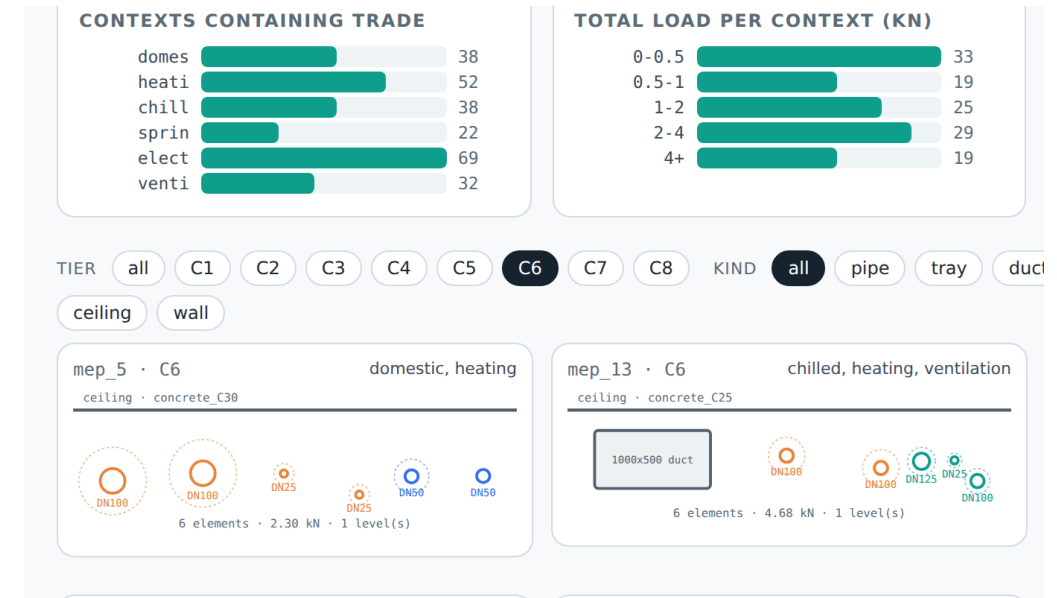
In: the parameters (tier C7, seed 0). Out: section A-A at the support, as the support designer receives it (scale 1:20), and the same run in 3D. Every section is a stored output of the released generator, with the Python call that reproduces it.

# Train on the generator, evaluate on the benchmark, report per tier

```
import crossmep.tasks as cm
data = cm.load("benchmark") # 1,000 contexts
cm.filter_contexts(data, pipes=2, trays=1)

from crossmep.evaluate import score, compare
score(my_results, data) # per tier, 95 % CI
compare(mine, baseline, data) # paired test
```

Loading, metrics and scoring need only the Python standard library; the generator needs NumPy. Command line: `python -m crossmep generate | validate | evaluate`.



Interactive gallery: filter the benchmark by tier, kind, trade and surface.

## Train

on the generator: unlimited seeds, a curriculum from C1 up to C8

## Evaluate

on the benchmark: 125 held-out contexts per tier, the same for every method

## Report

per tier with 95 % intervals, paired tests between methods

# One section at one support today; a checker and real data come next

## Scope

- One cross-section at one support: routing, branches and elevation changes are outside it; so is support spacing — the span is recorded per element, so a method can vary it
- Realism checked on sections of two open buildings, a residential duplex and a medical clinic; congested racks rest on coordination practice and standards
- Trade mix and tier composition are design choices, not survey results
- Synthetic: not for the structural design of real installations

## Next

- A public checker for support designs, with best-known costs from an exact solver on the small tiers
- The catalog as an input: generated catalogs, and tests on catalogs a method has never seen
- A real test set: sections from commercial projects with supports designed by engineers

**If you coordinate services or design supports**, open the gallery, filter for the scenes you know, and tell us what looks wrong. Every rule is one documented constant away from being changed.



# Support design now has open problems to learn from

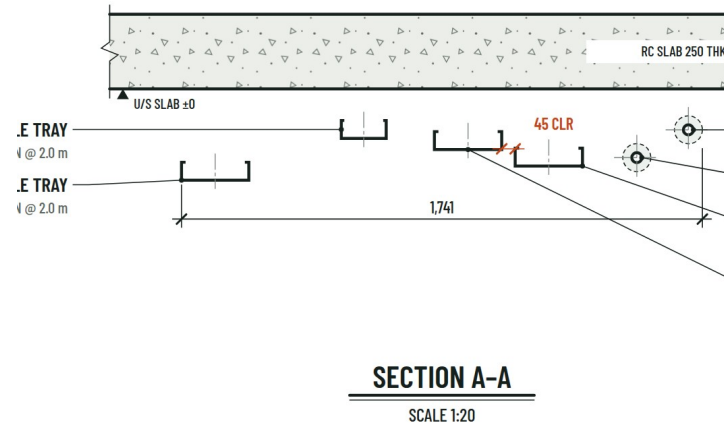
7,000 support-design problems · every constant sourced or declared  
pipe gaps checked on two open buildings · open data and code

Data CC BY 4.0, code MIT. Not for the structural design of real installations.

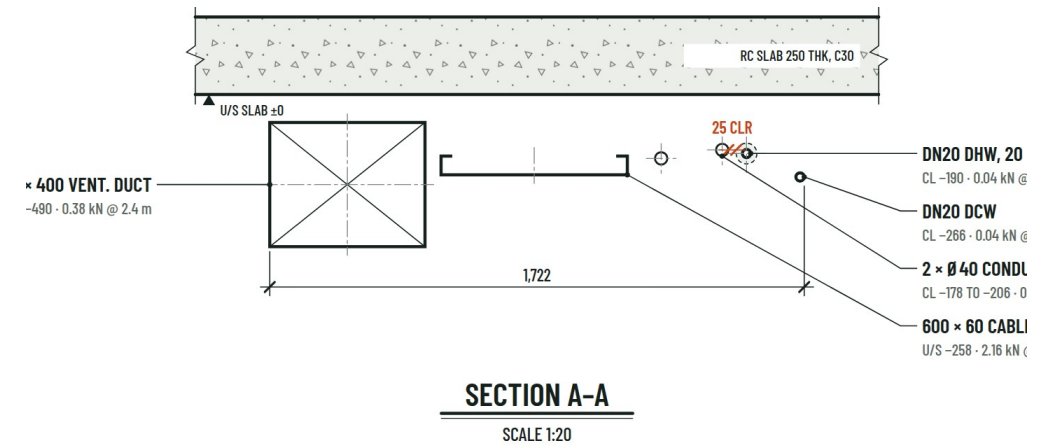


[github.com/lavinia-ped/CrossMEP\\_Dataset](https://github.com/lavinia-ped/CrossMEP_Dataset)  
lavinia@stanford.edu

# A wall with six pipes, and a slab with a duct, a tray and conduits



Tier C6, seed 0, section 5: six pipes fixed to a wall; offsets from the wall face, the closest clear gap 25 mm.



Exact mix: 2 pipes, 1 tray, 1 duct, 2 conduits on a slab; levels below the soffit and the load at the support.

# Median load rises from 0.30 kN at C1 to 2.04 kN at C8 as the gap narrows

| tier | contexts | median clear gap (mm) | median load (kN) | median width (mm) |
|------|----------|-----------------------|------------------|-------------------|
| C1   | 125      | —                     | 0.30             | 140               |
| C2   | 125      | 130                   | 0.24             | 431               |
| C3   | 125      | 89                    | 1.04             | 776               |
| C4   | 125      | 77                    | 0.94             | 1037              |
| C5   | 125      | 72                    | 1.22             | 1275              |
| C6   | 125      | 69                    | 1.36             | 1494              |
| C7   | 125      | 61                    | 2.24             | 1861              |
| C8   | 125      | 51                    | 2.04             | 2225              |

4,500 elements

2,488 pipes · 968 conduits · 559 trays · 485 ducts

32 mm

Wasserstein-1, generated to measured pipe gaps of the clinic  
(95 % CI 17–48)

–0.38 / +0.46

rank correlation of tier with the closest gap / with the load

# Against the clinic the generator is 32 mm off; the small duplex is 76 to 88

| pipe gaps             | pairs | sections | median (mm) | < 25 mm |
|-----------------------|-------|----------|-------------|---------|
| generated (benchmark) | 1,690 | —        | 216         | 0 %     |
| Clinic, Plumbing      | 797   | 3,833    | 203         | 2 %     |
| Duplex, MEP           | 74    | 204      | 177         | 23 %    |
| Duplex, Plumbing      | 41    | 112      | 114         | 32 %    |

- Sections every 250 mm; runs within 400 mm of height form a row; gaps between bare pipe surfaces, below 600 mm
- Robust: across 11 settings of the cut (spacing, row band, axis tolerance, run length) generated ↔ clinic stays at 31–41 mm
- Noise floor: the distance a perfect generator would still show at that sample size
- The duplex samples are small, hence the wide intervals

| Wasserstein-1 (mm)           | by length   | each pair once | noise floor |
|------------------------------|-------------|----------------|-------------|
| generated ↔ Clinic           | 32 (17–48)  | 18 (13–27)     | 8           |
| generated ↔ Duplex MEP       | 76 (61–103) | 97 (78–120)    | 23          |
| generated ↔ Duplex Plumbing  | 88 (62–132) | 104 (89–127)   | 32          |
| Clinic ↔ Duplex MEP          | 71 (54–101) | 88 (70–118)    | —           |
| Clinic ↔ Duplex Plumbing     | 85 (55–137) | 94 (77–121)    | —           |
| Duplex MEP ↔ Duplex Plumbing | 26 (24–124) | 25 (21–108)    | —           |

Intervals: 95 % bootstrap over contexts (generated) and pipe pairs (measured). Models: buildingSMART Duplex Apartment and Medical-Dental Clinic, CC BY 4.0; derived measurements and attribution in [verify/measured/](#).

# The diameter rule moves catalog coverage more than tier or split does

| tier | pipes | benchmark % (95 % CI) | population % (95 % CI) |
|------|-------|-----------------------|------------------------|
| C1   | 66    | 16.7 (8.3–26.6)       | 12.7 (10.7–14.7)       |
| C2   | 132   | 11.4 (5.0–18.4)       | 13.5 (11.8–15.4)       |
| C3   | 212   | 15.6 (10.5–21.1)      | 13.8 (12.4–15.2)       |
| C4   | 284   | 16.9 (11.8–22.1)      | 13.2 (12.0–14.4)       |
| C5   | 365   | 11.5 (7.3–16.0)       | 14.7 (13.5–15.9)       |
| C6   | 366   | 10.7 (7.2–14.5)       | 14.3 (13.3–15.4)       |
| C7   | 517   | 17.2 (13.4–21.1)      | 14.9 (13.9–16.0)       |
| C8   | 546   | 15.2 (11.7–19.1)      | 14.4 (13.5–15.4)       |

| attach diameter rule | bins $\pm 0$ mm | bins $\pm 0.5$ mm | bins $\pm 1$ mm |
|----------------------|-----------------|-------------------|-----------------|
| service (paper)      | 14.5            | 19.8              | 19.8            |
| bare                 | 14.5            | 22.0              | 22.0            |
| insulated            | 7.2             | 8.2               | 8.2             |

% of benchmark pipes attachable when both bin edges are widened by the stated amount. DN100 (114.3 mm) lies 0.3 mm above the 108–114 mm bin, so the headline moves with the diameter rule and the edge tolerance.

| split     | pipes  | % attachable (95 % CI) |
|-----------|--------|------------------------|
| train     | 12,590 | 14.4 (13.7–15.2)       |
| val       | 1,295  | 13.2 (11.2–15.6)       |
| test      | 1,200  | 13.8 (11.5–16.4)       |
| benchmark | 2,488  | 14.5 (12.7–16.2)       |

- Intervals: cluster bootstrap over contexts (2,000 resamples), because the elements of one context are correlated
- Population: 2,000 freshly generated contexts per tier; overall 14.3 % (95 % CI 13.8–14.7)
- Only size and capacity are tested, so coverage is an upper bound on what can be installed
- Best possible share with k sizes: exact optimum over the distinct attach diameters

# A context is stored as plain JSON: elements, sizes, loads, positions

```
{
  "context_id": "mep_9", "tier": "C2",
  "surface": {
    "kind": "ceiling", "substrate": "concrete_C25", "thickness_mm": 200.0
  },
  "n_elements": 2, "n_levels": 1, "total_load_kN": 0.76, "bundle_width_mm": 1527.0,
  "elements": [
    {
      "kind": "duct", "service": "duct", "trade": "ventilation", "shape": "rect",
      "label": "1000x500 duct", "width_mm": 1000.0, "height_mm": 500.0, "insulation_mm": 0.0,
      "load_kN": 0.72, "span_m": 2.4, "load_kN_per_m": 0.3002,
      "level": 0, "along_mm": -263.5, "out_mm": 340.0
    },
    {
      "kind": "pipe", "service": "domestic_cold", "trade": "domestic", "shape": "round",
      "label": "DN20", "width_mm": 26.9, "height_mm": 26.9, "insulation_mm": 0.0,
      "load_kN": 0.04, "span_m": 2.1, "load_kN_per_m": 0.0189,
      "level": 0, "along_mm": 750.0, "out_mm": 341.3
    }
  ]
}
```

- Units: millimetres, kilonewtons, metres (span)
- Widths are bare; insulation\_mm is per side
- $\text{load\_kN} = \text{load\_kN\_per\_m} \times \text{span\_m}$
- along\_mm, out\_mm: position along and out from the surface
- level: the generative row (metadata)